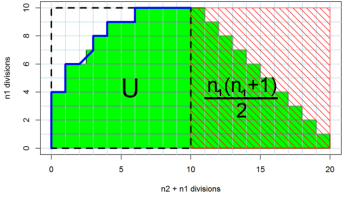
# Introduction to Machine Learning

# Evaluation
# AUC & Mann-Whitney-U Test

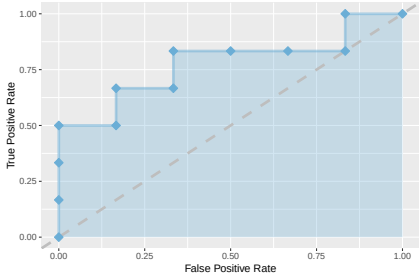



**Learning goals**

- Understand the rank-based nature of AUC
- See the connection between AUC and Mann-Whitney-U statistic

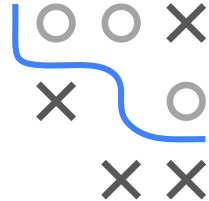
# AUC AS A RANK-BASED METRIC

- The AUC metric is intimately related to the **Mann-Whitney-U test**, also known as **Wilcoxon rank-sum test**.
- This connection is best understood viewing the AUC from a slightly different angle: it is, in effect, a **rank-based** metric.
- Recall that, constructing the ROC curve, we count TP and FP.




- The AUC abstracts from the actual classification scores and considers only their rank.

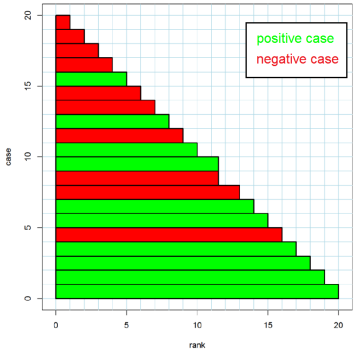
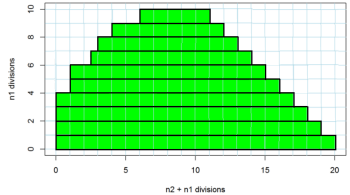
# MANN-WHITNEY-U TEST



- The Mann-Whitney-U test is a **non-parametric hypothesis test** on the difference in location between two samples $X_1$, $X_2$ of sizes $n_1$ and $n_2$, respectively.
- Under the null, $X_1$ and $X_2$ follow the same (unknown) distribution $\mathbb{P}$, and for any pair of observations $x_{1,1} \in X_1$, $x_{2,1} \in X_2$ drawn at random from $\mathbb{P}$, the following statement holds: $\mathbb{P}(x_{1,1} \in X_1) > \mathbb{P}(x_{2,1} \in X_2) = \mathbb{P}(x_{1,1} \in X_1) < \mathbb{P}(x_{2,1} \in X_2) = 0.5$.
- The test statistic estimates the probability of a random sample from $X_1$ ranking higher than one from $X_2$ ($R_1$ denoting the sum of ranks of the $x_{1,i}$):

$$U = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbb{I}[x_{1,i} > x_{2,j}] = R_1 - \frac{n_1(n_1+1)}{2}$$

- For large samples, $U$ is approximately normally distributed.

# AUC & MANN-WHITNEY-U TEST

- We can directly interpret the AUC in the light of the U statistic.
- In order to see this, plot the ranks of all the scores as a stack of horizontal bars, and color them by label.
- Next, keep only the green ones, and slide them horizontally to get a nice even stairstep on the right edge: